

Sedentary time predicts an increased annual risk of transitioning from healthy to burnout-related coping patterns, while physical activity does not

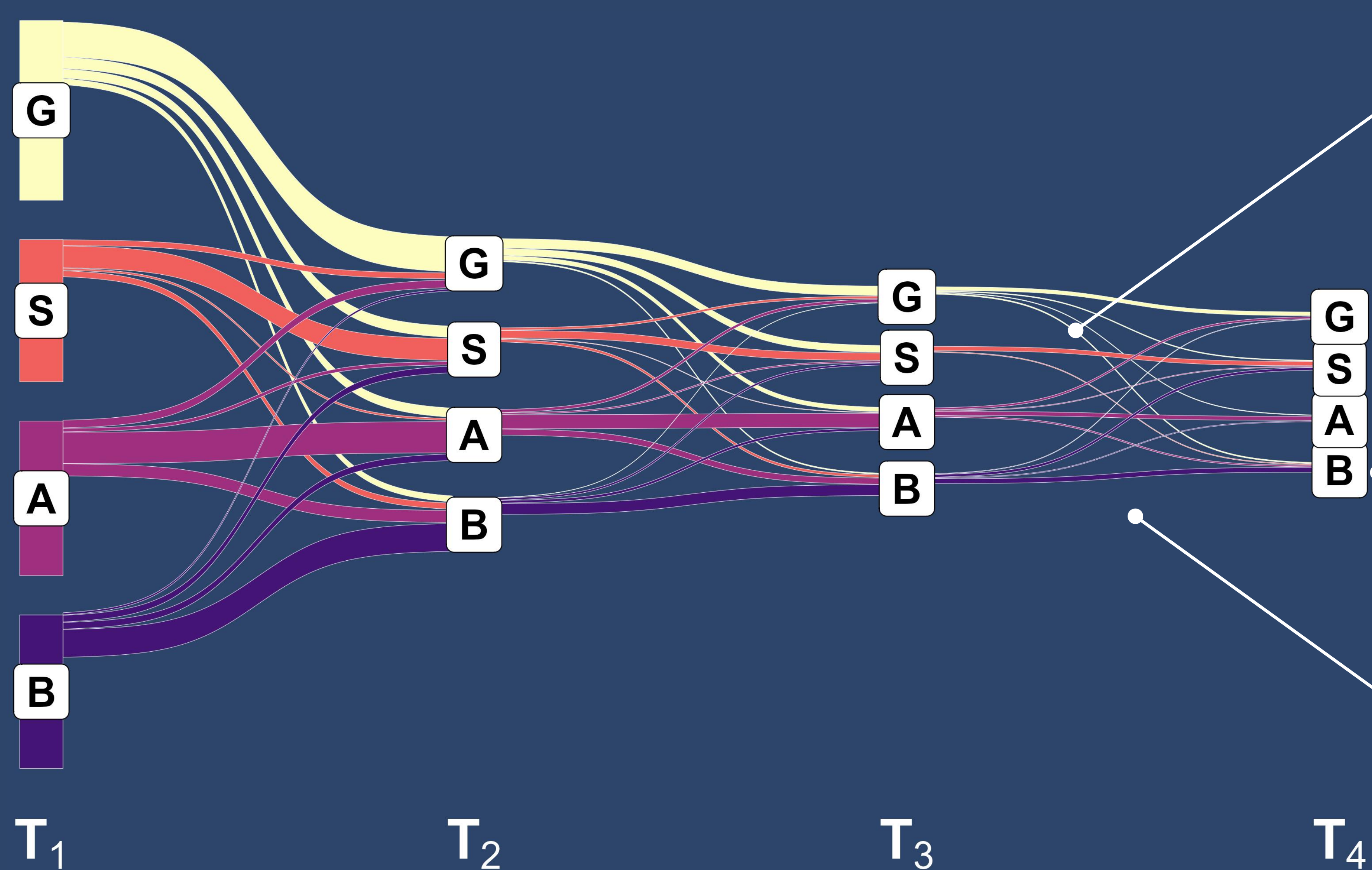


Figure 1. Proportion of students in each coping pattern at T_1 – T_4 (person-centered time points aligned with each participant's first participation between 2022 - 2025), indicated by node widths, and their proportional transitions across consecutive years, indicated by flow widths.

Results

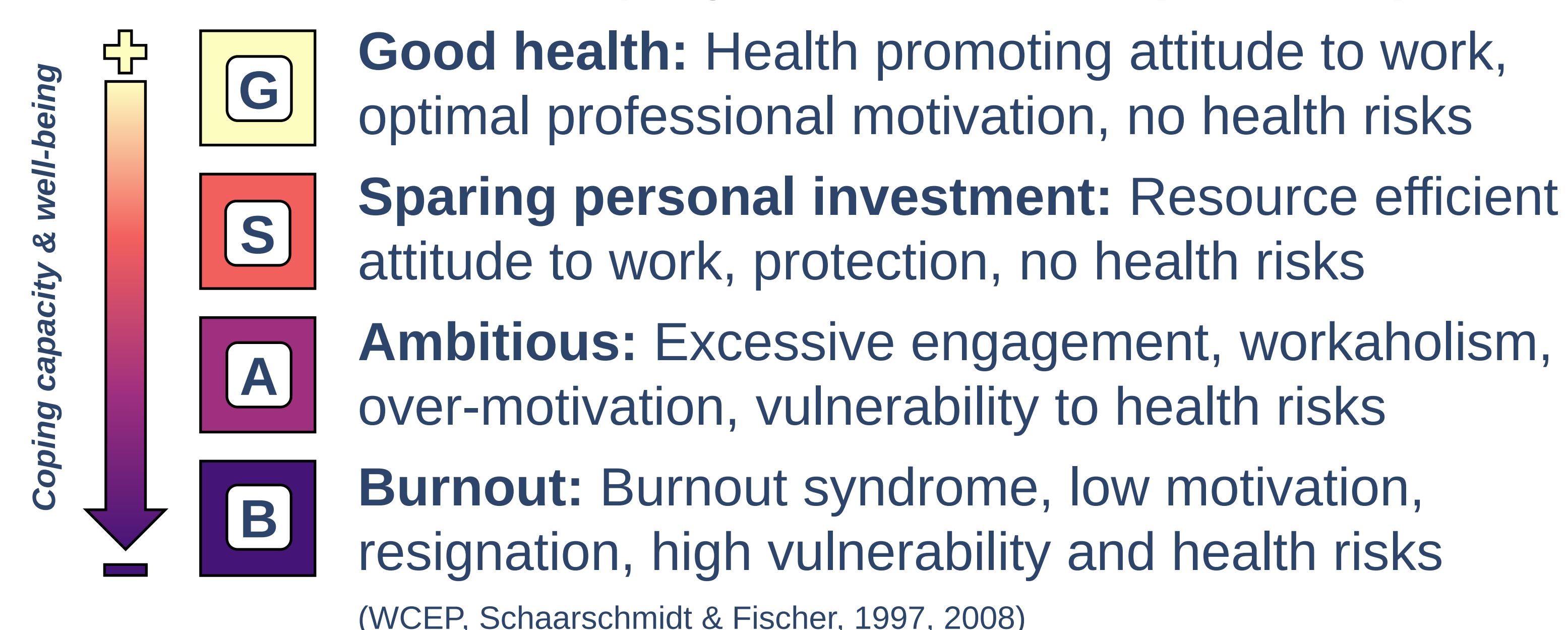
- 1 Sedentary time**
Higher sedentary time predicts transitions from healthy (G) to burnout-related coping (B) within one academic year ($HR = 2.30$, 95% CI: 1.15 – 4.59).
- 2 Transition probability**
The probability of B (Burnout) being the next patterns after G (Good health) was 3% (95% CI: 1% – 22%) for low (-1 SD) and 16% (95% CI: 6% – 38%) for high ($+1$ SD) sedentary time.
- 3 Physical activity**
Physical activity was not significantly associated with transition hazard ratios between coping patterns ($HRs = 0.77 - 1.29$, all 95% CIs included 1).

Research question

How are physical activity (PA) and sedentary time (ST) associated with transitions between academic coping behavior and experience patterns in students over time?

Method

Outcome: Academic coping behavior and experience patterns



Design

Longitudinal cohort survey study (2022 – 2025)

$N_{T1} = 785 \rightarrow N_{T2} = 677 \rightarrow N_{T3} = 341 \rightarrow N_{T4} = 162$

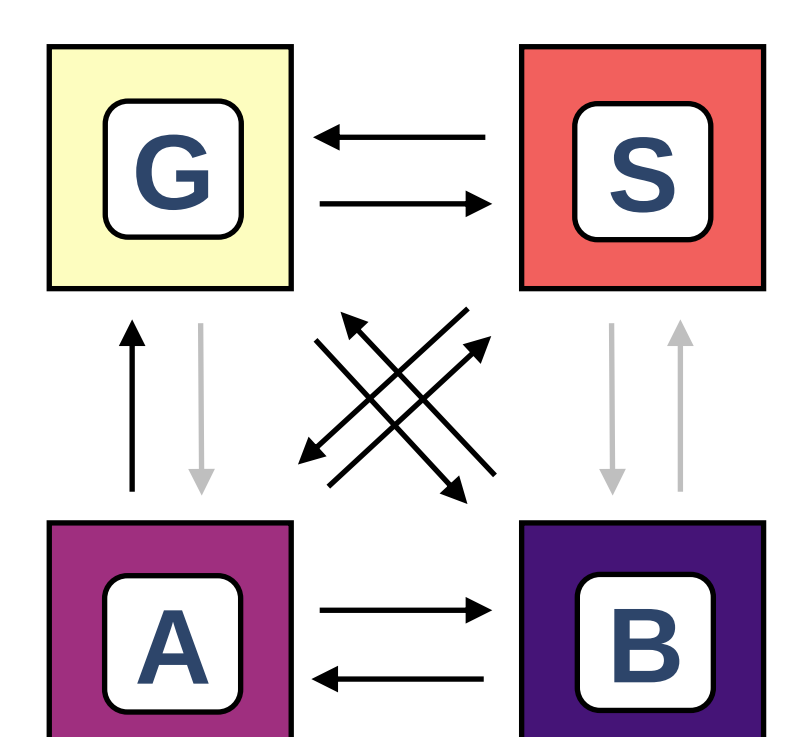
Predictors

PA: Moderate to vigorous intensity PA in metabolic equivalent of task minutes per week (IPAQ, Craig et al., 2003)

ST: Daily sitting time in minutes (IPAQ, Craig et al., 2003)

Data analysis

Continuous-time multi-state Markov models were applied to estimate the effects of PA and ST on transition probabilities between patterns over time.



When Sitting Still Leads to a Standstill: The Role of Physical Activity and Sedentary Time in Coping with Academic Stress

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